

ZINAL RAVAL

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Professional Summary

Motivated and innovative Artificial Intelligence and Data Science student with hands-on experience in machine learning, deep learning, NLP, and robotics. Proven track record in building end-to-end AI solutions, data pipelines, and real-time intelligent systems. National hackathon winner and SSIP India funding recipient for robotics projects. Passionate about solving real-world problems using AI and creating meaningful tech solutions.

Education

Bachelor of Technology in Artificial Intelligence & Data Science (Minor: Robotics)

A.D. Patel Institute of Technology, Anand, Gujarat (CVM University) - Expected Graduation: 2025

Higher Secondary Education

P. P. Savani Vidyabhavan, Surat, Gujarat — Completed: 2021

Technical Skills

- Programming Languages: Python, SQL, C/C++, HTML, CSS
- Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, LangChain
- Tools & Platforms: FastAPI, Arduino IDE, Selenium, FFmpeg, Git, VS Code, Jupyter Notebook
- Databases & Search: FAISS, ChromaDB
- Other Technologies: OCR (Tesseract, PyMuPDF, pdf2image), Hugging Face, Groq, RAG, SentenceTransformers

Projects

AI-Powered Autonomous Research & Decision System (AR-DMS)

- Built a FastAPI-based AI research assistant to process PDFs, web content, and images using OCR, semantic search (FAISS), summarization, and NLP tools (LangChain, Hugging Face).

Text-to-SQL Generator (CodeT5 + RAG)

- Fine-tuned a CodeT5 model with RAG for generating SQL queries from natural language, including multi-table joins and schema-aware responses.

Material Estimation from Architectural PDFs

- Used Groq + LLaMA 3 with PyMuPDF and pdfplumber to extract structured room-wise data and generate CSV-based material takeoff reports.

AI-Powered GPT-4 Text Rewriting Tool

- Rewriting system using GPT-4 (OpenRouter) to humanize technical text; validated tone and semantic similarity with SBERT.

LinkedIn Scraper & Business Insights Tool

- Automated professional data scraping using Selenium + SerpAPI with AI-based business classification, proxy rotation, and JSON output.

HistoriAI – Multilingual AI Travel Assistant [text to speech – MMS]

- Tourism chatbot powered by LLaMA 3, LangGraph, and ChromaDB supporting English, Gujarati, and Hindi with RAG-based search.

Robotic Arm (SSIP Funded)

- Arduino-based robotic arm for automated pick-and-place tasks with real-time control and precision movements.

Robotic Nurse (SSIP Funded – Ongoing)

- Prototyping an autonomous robot for hospitals that delivers medication and monitors vital signs using embedded sensors.

Experience

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| 1. AI/ML Developer | Prodev Infotech Feb 2025 - Present |
| 2. AI/ML Intern | Technoyuga Soft Pvt. Ltd. May 2024 – May 2024 |
| 3. Machine Learning Intern | Prodigy InfoTech Jan 2024 – Feb 2024 |
| 4. Data Science Intern | Oasis Infobyte Feb 2024 – Mar 2024 |
| 5. Core Team Member | MooRopan INDIA PVT. LTD. Oct 2022 – May 2023 |
- Assisted in the development of an AI-based device for mastitis detection in cows.

Achievements

- Winner – **Reckon 4.0** National Hackathon: 1st Place for developing an AI solution with real-world impact.
- SSIP India Funding Recipient: Secured funding for **Robotic Nurse** and **Robotic Arm** projects.
- SSIP Hackathon 2022 Regional Selection: Recognized for innovative tech solutions.